TASK 3: Automated System Design – Automated Packaging System

1. Introduction

An automated packaging system places products into boxes, seals them and counts the output without manual handling. This design uses an Arduino Uno as the controller, two infrared (IR) sensors for detection, a DC gear motor to drive the conveyor belt, and servo motors for pushing and sealing. It is low-cost, easy to build as a project, and demonstrates the same sensing-decision-actuation pattern used in industrial packaging lines.

2. Components Used

Component	Quantity	Function
Arduino Uno (ATmega328P)	1	Main controller; runs the control program
IR obstacle sensor	2	IR-1 detects an empty box at the station; IR-2 detects a product
DC gear motor + belt	1	Moves empty boxes along the conveyor
L298N motor driver	1	Switches the 12 V motor and controls its speed with PWM
Servo motor (SG90/MG90)	2	Servo 1 pushes the product into the box; Servo 2 presses / seals the lid
16×2 I2C LCD	1	Shows the count of packed boxes and the status
Push buttons	2	START and STOP
Buzzer + red and green LEDs	1 + 2	Signals running, stopped, or no product
12 V DC supply	1	Powers the motor driver; Arduino also takes power from it

Table 5: Components

3. Working Principle

When the START button is pressed, the Arduino switches the conveyor motor on through the L298N driver. Empty boxes travel along the belt until IR-1 detects a box at the packing station, and the controller stops the belt.

The controller then waits for IR-2 to confirm that a product is present on the feeder. Once the product is detected, Servo 1 pushes it into the box, Servo 2 performs the sealing or pressing action, and the counter on the LCD increases by one. The conveyor then restarts and moves the filled box forward, and the next empty box arrives. If no product arrives within a set time, the buzzer and red LED warn the operator. Pressing STOP, or running out of boxes, ends the cycle, and the system returns to idle.

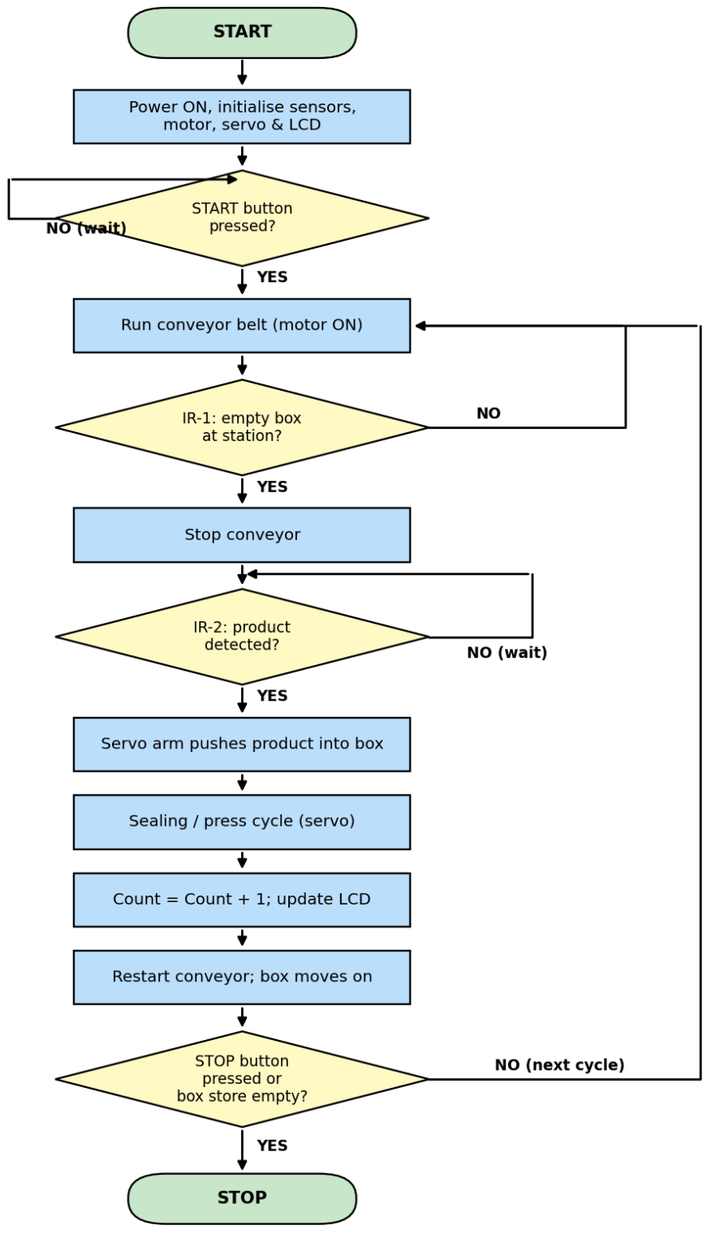


Figure 3: Flowchart of the automated packaging system

4. Circuit (Wiring) Diagram

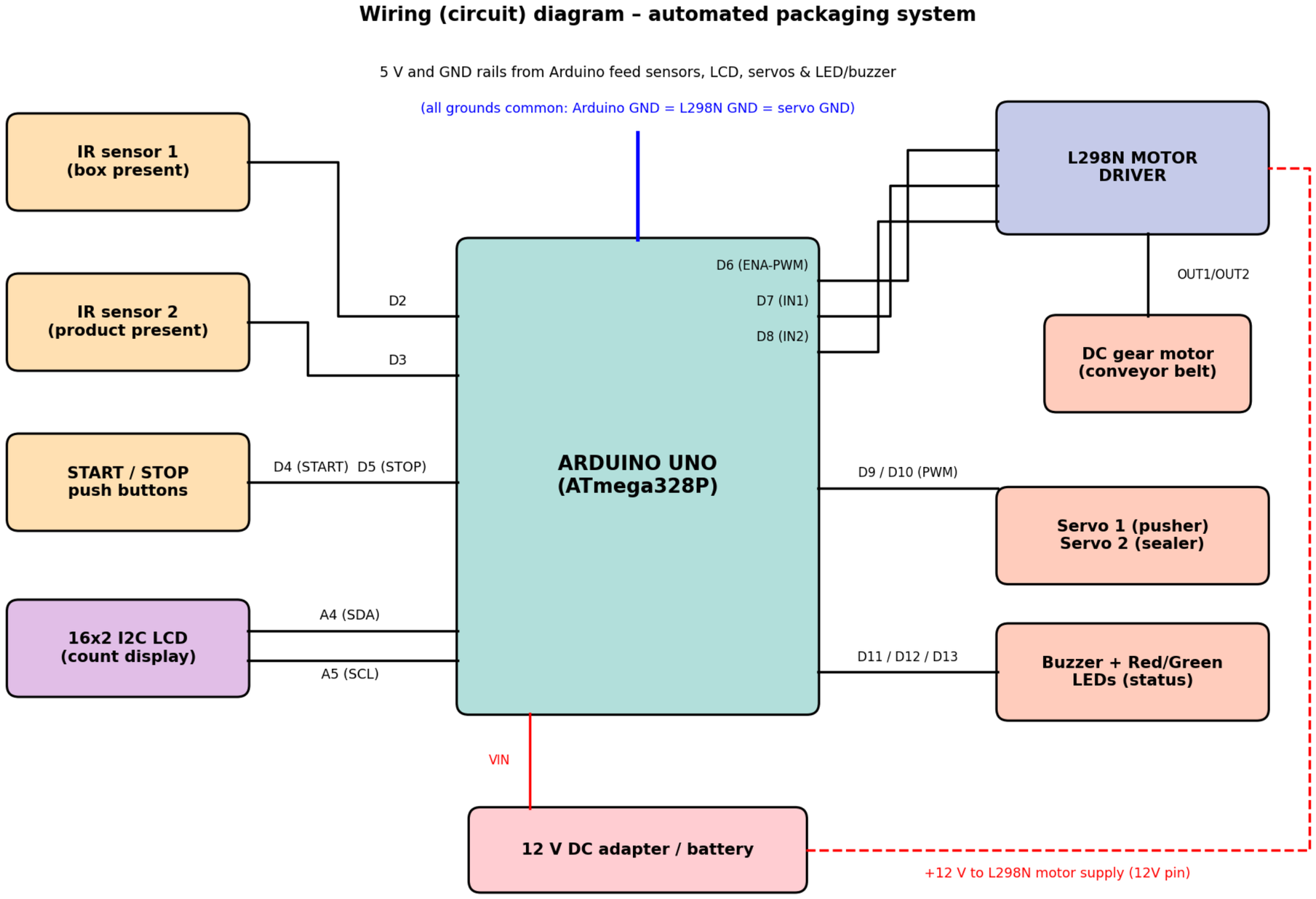


Figure 4: Wiring diagram – sensors, driver, servos and display connected to the Arduino Uno

Device	Arduino pin	Remark

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IR sensor 1 (box present)	D2	Digital input, output LOW on detection
IR sensor 2 (product present)	D3	Digital input
START / STOP buttons	D4 / D5	With INPUT_PULLUP, other side to GND
L298N ENA / IN1 / IN2	D6 (PWM) / D7 / D8	Speed and direction of the conveyor motor
Servo 1 / Servo 2	D9 / D10	PWM signal; powered from 5 V supply rail
Buzzer, red LED, green LED	D11 / D12 / D13	LEDs through 220 Ω resistors
LCD (I2C)	A4 (SDA) / A5 (SCL)	5 V and GND; address typically 0x27

Table 6: Pin connections

Note: the motor and the servos should not be powered from the Arduino's 5 V pin when they draw heavy current; use a separate 5 V supply for the servos and connect all grounds together.

5. Advantages and Applications

- **Advantages:** low cost, consistent packing speed, reduced manual labour and contamination, an accurate count of packed boxes, and easy modification because the logic is in software.
- **Applications:** small food and snack units, pharmacy tablet packing, courier and parcel sorting, and bottling or cartons in small factories and teaching laboratories.

6. Conclusion

The designed system automates the detect–stop–pack–seal–count cycle using simple sensors and actuators controlled by an Arduino. With stronger motors, more sensors and a PLC, the same logic scales to industrial packaging lines.